

Problem

Submissions

Leaderboard

Discussions

The 3N+1 sequence is generated using the following rules: If n is even, the next number is $n / 2$. If n is odd, the next number is $3 \times n + 1$. The sequence continues until it reaches 1. The cycle length of a number is the total number of terms in the sequence, including the starting number and 1.

Given two integers i and j, determine the maximum cycle length among all integers between i and j (inclusive).

If $i > j$, swap them before processing.

Input Format

A single line containing two space-separated integers:

i j

Constraints

$1 \leq i, j \leq 1,000,000$

Output Format

Print three integers:

i j maximum_cycle_length

using the original values of i and j.

Sample Input 0

1 10

Sample Output 0

1 10 20

Change Theme

Language Python 3

↺

⋮

```
1 # Enter your code here. Read input from STDIN.
2 i, j = map(int, input().split())
3
4 original_i = i
5 original_j = j
6
7 if i > j:
8     i, j = j, i
9
10 max_cycle = 0
11
12 for n in range(i, j + 1):
13     x = n
14     count = 1
15
16     while x != 1:
17         if x % 2 == 0:
18             x = x // 2
19         else:
20             x = 3 * x + 1
21
22         count += 1
23
24     if count > max_cycle:
25         max_cycle = count
26
27 print(original_i, original_j, max_cycle)
28
```

Line: 28 Col: 1

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?

✔ Test case 0

✔ Test case 1

✔ Test case 2

Compiler Message

Success